FS Onboarding : Power Electronics

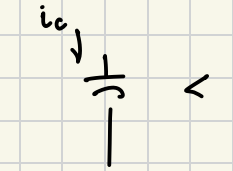
Activity 1: Initial Learning

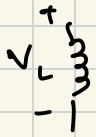
Analog Devices + Ckts

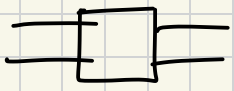
→ $V = IR$

→ $V_2 = V_1 \left(\frac{R_2}{R_1 + R_2} \right)$

→ KVL, KCL



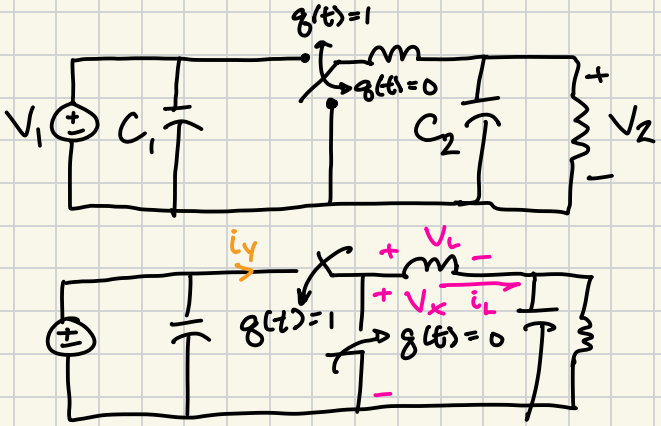


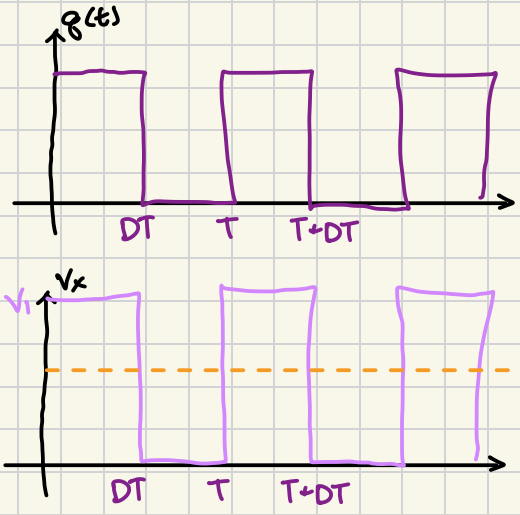


if $n = 1$

$P_{in} = P_{out}$

Intro to DC/DC





$V_x - V_L - V_2 = 0$

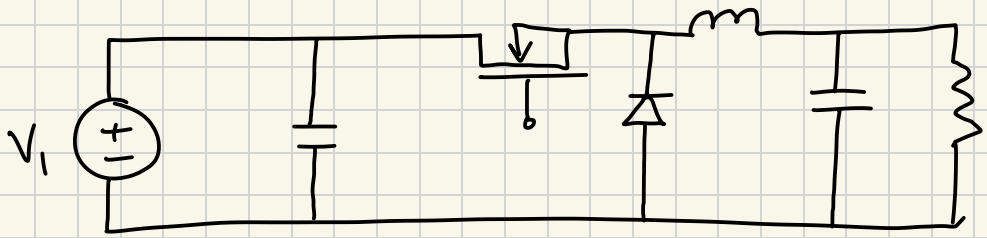
$\langle V_x \rangle - \langle V_L \rangle - \langle V_2 \rangle = 0$ (0 in PSS)

$\langle V_2 \rangle = \langle V_x \rangle$

$V_2 = DV_1$

$i_L(t) \approx I_L$, $v_1(t) \approx V_1$

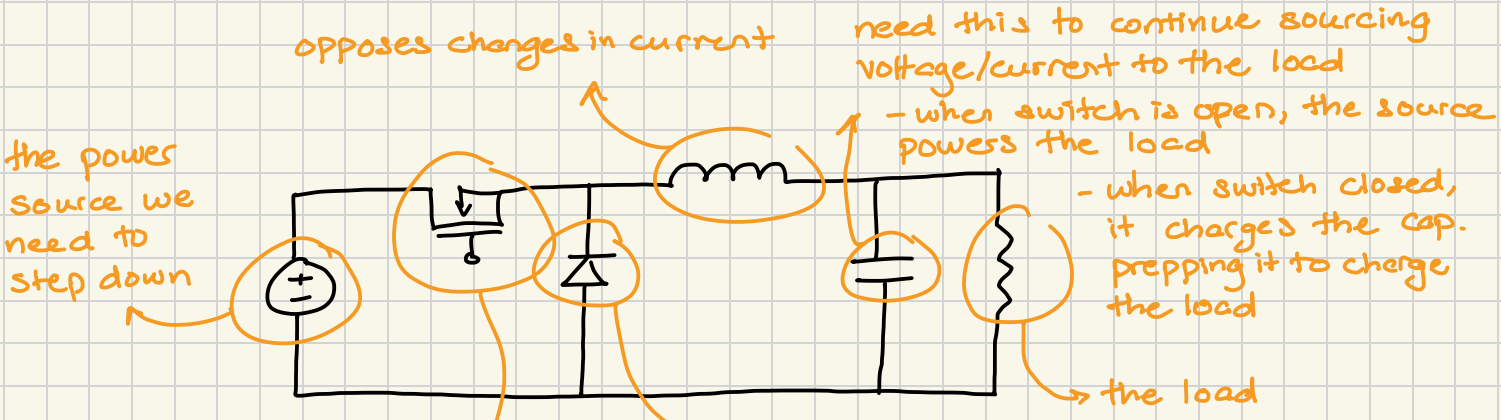
$v_2(t) \approx V_2$



"Buck Converter"

↳ steps input voltage down

essentially opposes changes in voltage



"Asynchronous Buck Converter"

due to the diode,
replacing it w/ a MOSFET would make it synchronous

MOSFET
- drain to source voltage raining
- current rating
- power dissipation rating
- thermal characteristics
- duty cycle

INDUCTOR
- inductance
- RMS current rating
- saturation current rating
- equivalent series resistance

DIODE
- reverse voltage breakdown
- forward voltage
- current rating
- power dissipation rating

$$T_{switching} = T_{on} + T_{off}$$

$$D\,[\text{duty cycle}] = \frac{T_{on}}{T} \;;\; [0 < D < 1]$$

$$V_{\substack{out \\ avg}} = \frac{V_{in} * T_{on}}{T} = V_{in} D$$

Activity 2: Component Selection

☑ Buck IC

- purpose: adjusts duty cycle of switches & works w/ the feedback loop
- MUST output enough current to supply all boards (anything 2A+ is good)
  - ↳ going too high in current can have downsides however
- MUST understand what each pin does, what kind of internal structure it has, & the reasoning behind all of it

☐ Inductors

ensure I_{sat} exceeds I_{peak} by at least 20-30% under worst case operating temps

- CHECK FOR: saturation current
  - make sure it doesn't oversaturate @ the output current you're expecting,
    else its inductance will drop so low that the output ripple will be extremely large

☐ Capacitors

- they don't like staying @ the specific capacitance they were designed to be at

  Instead, whenever voltage is applied to it, it will drop by some percentage ("DC biasing"), shown in its datasheet
- CHECK FOR: ensure capacitances will be what you're expecting at the expected output voltage
  [hint: output capacitors are used to filter out noise]

to account for DC biasing, we need to choose appropriate voltage rating

→ minimum rated voltage $\geq 2 \times V_{out}$

$\geq 2(5) \geq$ 10V

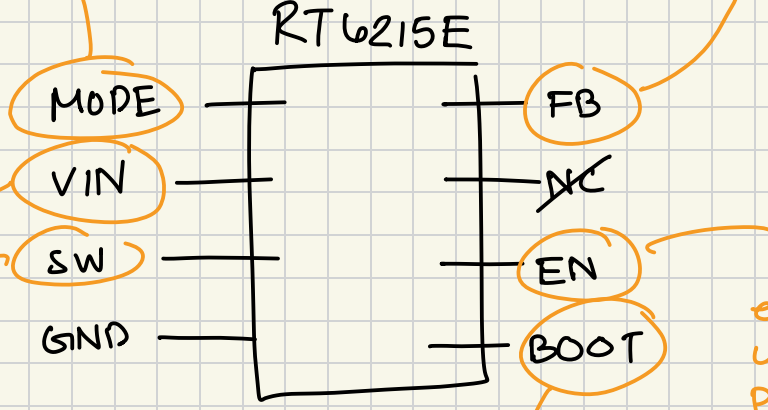
lets you choose how IC behaves when not pulling a lot of current

↳ Continuous Conduction Mode (CCM)
internal switches will toggle @ 500KHz no matter what output current is
↳ use for low voltage ripple, less noise, & easy filtering

↳ Pulse Skipping Mode (PSM)
saves energy over keeping constant switching freq.
↳ use for light loads

monitors V_{out} through resistive divider to keep V_{out} stable at 5V

supports 4.5-24V input voltage
NEEDS large ceramic capacitor



enables the buck, useful becuz it protects the IC & can serve as a power conservation mechanism if need be

pin that acts like the common line/trace that carries the current flowing out to the inductor & communicates the voltage difference to whicheever position the switch is on (ie. on/up 13.5V or off/down 0V)

helps the top switch fully switch ON as it needs a voltage higher than what it's outputting to stay in the up position



Inductor → for RT6215E, to compromise between size, efficiency, & transient response, we set an inductor ripple current (ΔI_L) about 20-50% of desired full output load current

$$L = \frac{V_{out}(V_{in}-V_{out})}{V_{in} f_{sw} \Delta I_L} = \frac{5(18.5-5)}{13.5(500\times10^3)(0.7)} = 8.995\,\mu H$$

↳ 35% of max output current [2A]

$$\Delta I_L = \frac{V_{out}(V_{in}-V_{out})}{V_{in} f_{sw} L} = \frac{5(13.5-5)}{13.5(500\times10^3)(9\times10^{-6})} = 0.6995\,A$$

↳ around 35% of max o/p current!!

$$I_{L(peak)} = I_{out(max)} + \frac{1}{2}\Delta I_L = 2 + \frac{0.6995}{2} = 2.34\,A$$

$$I_{L(valley)} = I_{out(max)} - \frac{1}{2}\Delta I_L = 2 - \frac{0.6995}{2} = 1.65\,A$$

OR use 8.2 μH

saturation current target (20-30% of $I_{L(peak)}$) ⇒ 2.808A – 3.042A

Input Capacitor → for RT6215E, the voltage rating MUST be greater than the max. input voltage

$$I_{RMS} = I_{out(max)}\left(\frac{V_{out}}{V_{in}}\right)\sqrt{\frac{V_{in}}{V_{out}} - 1}$$

$$= 2\left(\frac{5}{13.5}\right)\sqrt{\frac{13.5}{5} - 1} = 0.965\,A$$

- consider 22μF & 0.1μF in parallel? (recommended in DS)

- stick w/ ceramic caps
- be careful using caps at inputs of ckts supplied by wall adapter or supply connected by long & thin wires
- choose caps rated @ higher temps than required

Output Capacitor → for RT6215E,

- consider two 22μF caps in parallel ⇒ $C_{out} = 44\,\mu F$ (recommended in DS)

$C_{eff} = 44\,\mu F(1-0.30)$

$= 30.8\,\mu F$ ← the "actual" Cout considering DC biasing

$$C_{eff} = \frac{\Delta I_L}{8 f_{sw} \Delta V_{out}}$$

Resistive Divider → for RT6215E

use resistors w/ 1% or better tolerance

→ fix $R_2 = 20k\Omega$, $V_{ref} = 0.791\,V$

→ $R_1 = \dfrac{R_2 (V_{out} - V_{ref})}{V_{ref}} = \dfrac{(20\times10^3)(5 - 0.791)}{0.791}$

$= 106422\,\Omega \approx 105\,k\Omega$

Voltage Ripple

$V_{ripple} = V_{ripple(ESR)} + V_{ripple(C)} \Rightarrow V_{ripple} = 7.38mV$

$V_{ripple(ESR)} = \Delta I_L \times R_{ESR} = 0.768A \times 0.0015\Omega = 1.15\,mV$

$V_{ripple(C)} = \dfrac{\Delta I_L}{8 C_{out} \cdot f_{sw}} = \dfrac{0.6945}{8(30.8\times10^{-6})(500\times10^3)} = 6.23\,mV$

$V_{sag} = \dfrac{L(\Delta I_{out})^2}{2C_{out}[V_{in(min)} D_{max} - V_{out}]}$

$V_{sacr} = \dfrac{L(\Delta I_{out})^2}{2 C_{out} V_{out}}$

make sure it's 1-5% of the output voltage!!!

$7.38\,mV \ll 1\%\, V_{out}$ ($50\,mV$) ✓

Cost

[check Onboarding sheet]

Activity 3: Schematic Design